

Group Project

ARI 2129 - Principles of Computer Vision for AI

Project Title: Computer Vision Learning Pack

Deadlines: 18th May 2026

Group Size: Groups between 4 and 5 students (Equal effort will be assumed)
Register here: <https://forms.gle/ehfdR3CuwrYuKzzv7>

Maximum Marks: 100 (60% of the entire study unit)

Plagiarism and Authenticity:

- No plagiarism will be tolerated. All sources must be adequately referenced in the code and documentation.
- Generative AI (GenAI) is allowed as long as it is documented, in line with the instructions in this documentation, and reported in the Generative AI Journal.
- The use of generative AI needs to be declared and documented.
- A random number of students will be asked to participate in impromptu interviews to assess their knowledge of the submitted content. Any information in the submitted documentation that is not adequately explained by the students and lacks its respective sources will be reported as plagiarism immediately.

Learning Pack Assignment Brief:

"If you want to master something, teach it." — Richard Feynman

This is a **team assignment**. Each team will be assigned one core building block of computer vision. Your mission is to transform that building block into a high-quality **Learning Pack** that your peers can learn from independently. Teams consist of 4 or 5 students. Each team must cover a unique topic. Topic registration is handled via the designated registration tool. Once a topic is taken, it is no longer available.

Deliverables

You are to produce and submit the following six components.

1. Study Notes and Quick Reference

A written guide that bridges the gap between technical theory and intuitive understanding. Notes must be between **8 and 12 pages**.

The notes should cover:

- The why and how of your topic. Use analogies to make the abstract concrete.
- A technical section covering the essential algorithms or mathematical foundations.
- At least one worked numerical example showing step-by-step computation.
- Real-world examples and applications of the technique.

The notes must also include the following two reference sections, which may be placed at the end or integrated where appropriate.

Quick Reference: A cheat sheet summarising key concepts, equations, functions, and parameter ranges for your topic. Include a glossary of common misconceptions and concise definitions of key terminology. Think of this as something a student could consult at a glance while working or studying

Key Papers: A minimum of four papers directly relevant to your topic. For each paper, provide a full citation, a direct URL, a summary of approximately 100 words, and a sentence or two explaining its relevance. The PDF of each paper must be included in your submission inside a `further_reading/` folder. Papers that cannot be found, do not exist, or do not match the citation provided will be penalised. This requirement exists because AI tools sometimes generate plausible-sounding but fictitious references. You are expected to verify every paper you cite by locating and reading it.

Submission format: DOCX (editable) **and** PDF, both required in the repository.

2. Adversarial Quiz

A 10-question assessment designed to test deep comprehension rather than rote memorisation.

- Multiple-choice questions in which the incorrect answers (distractors) are based on common misconceptions or subtle edge cases.
 - Create the quiz in Google Forms. Each question must include answer feedback.
 - Make the Google Form viewable by anyone with the link (no login required).
 - Submit a separate document (DOCX and PDF) with the link to the Google Form explaining exactly why the correct answer is right and why each incorrect option is a plausible pitfall. Maximum 2 pages.
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3. Presentation Slides

A deck of 10 to 20 slides for your 10-minute in-class team presentation. Slide count is flexible. What matters is that your team can deliver the material clearly and comfortably within the time limit.

- Avoid text-heavy bullet points. Use diagrams, flowcharts, and images to tell the story of how your technique works.
- One slide must specifically address failure cases or limitations.
- One slide must address when not to use this technique and what alternatives exist.
- Include at least one real-world case study or application.

Submission format: PPTX (editable) and PDF both required in the repository.

4. Interactive Principle Simulator

A functional, interactive tool built in Python (Streamlit or Gradio) or a web-based framework.

- Users must be able to adjust parameters and see the visual output update
 - Code must be clean, commented, and executable by your peers with no additional setup beyond the provided requirements file.
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5. Annotated Code Walkthrough

A Jupyter notebook that walks through your technique in a narrative, tutorial style. The depth and scope of this deliverable should reflect the nature of your topic.

- Do not simply demonstrate a single library call. Walk through a meaningful pipeline, explain each step, show intermediate results, and demonstrate the effect of changing key parameters.
 - Code and explanation must be interleaved. Think of it as a technical blog post, not a script with comments.
 - Use OpenCV and other standard libraries as appropriate.
 - All code must be clean, well-structured, and executable.
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6. AI Usage Journal

This assignment shifts the question from "Did you use AI?" to "How did you use AI?" while adopting a "show your working" approach. You are expected to use AI tools and to be honest and specific about how you did so.

The journal has a maximum of **2 to 4 pages per individual, compiled into a single journal**, and covers three sections.

Section 1: Tools declared. List every AI tool your team used, with the version where known. One short sentence per tool explaining what you used it for.

Section 2: Notable prompts. Select 4 to 6 prompts that best support your answers in Section 3. For each, include the prompt text and a brief summary of the response. Focus on moments that genuinely changed something in your work, not routine interactions.

Section 3: Five questions. Answer each of the following in 50 to 100 words.

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?
2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?
3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?
4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?
5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

Answers are assessed on honesty and specificity. A journal with no friction, no errors, and no wasted time is not credible. It is ok to be raw in the journal and share/reflect on your use of AI. This is the level of transparency any employer would expect from you. Your notable prompts in Section 2 should provide direct evidence for your answers in Section 3.

Assessment Structure

Weighting

Component	Weight
Study Notes (including Quick Reference and Key Papers)	20%
Simulator	20%
Annotated Code Walkthrough	15%
Adversarial Quiz	10%
Presentation Slides	10%
Live Presentation	15%
AI Usage Journal*(individual mark)	10%
Total	100%

All team members receive the same mark.

Presentation Format

Presentations take place across the last two lecture sessions. Each presentation lasts **10 minutes**, followed by approximately **5 minutes** for questions directed to any team member. Your slides (deliverable 3) are your presentation materials. Presentation order follows the block structure. Teams covering foundational topics present first.

Submission Requirements

Your submission must be a **public Git repository** (GitHub, GitLab, or equivalent) with a clean commit history that reflects genuine iterative development. A single commit containing all files suggests the repository was created at the last minute rather than used as a working tool throughout production. Commit regularly and meaningfully as you build each deliverable. Submit the repository link and a zip archive on the VLE.

Repository Structure

Your repository must follow this structure exactly and reflect the deliverables.

```
/
├─ README.md
├─ requirements.txt
├─ study_notes.docx
├─ study_notes.pdf
├─ quiz_with_rationale.docx
├─ quiz_with_rationale.pdf
├─ quiz_link.txt
├─ slides.pptx
├─ slides.pdf
├─ walkthrough.ipynb
├─ ai_journal.pdf
├─ simulator/
│   └─ app.py
│   └─ requirements.txt
│   └─ README.md
└─ further_reading/
    └─ *.pdf
```

Simulator Requirement

The simulator must be fully functional directly from the repository. A reviewer must be able to clone your repository, run `pip install -r requirements.txt`, and launch the simulator with a single command as documented in your README. If it does not run, it will not be marked. Your commit history matters. All code must be clean, well-commented, and executable. Variable names, function names, and comments must be in English.

Topic Allocation

Each team covers one of the following topics. Topics are organised into seven thematic blocks that reflect their conceptual relationships. Topic selection is to be carried out through this link:

<https://docs.google.com/spreadsheets/d/1g2vOl9y5kXohLSYcCrLy0a9364xE5ZFJv4MYwmRY380/edit?usp=sharing>

#	Block	Topic	Core Focus
1	Image Data	Colour models and colour spaces	RGB, HSV, YCbCr, LAB. Conversion equations, why each exists, what breaks when you use the wrong one.
2	Image Data	Image formats and compression	Lossless vs lossy, JPEG DCT pipeline, PNG prediction filters, compression artefacts and when they matter.
3	Image Data	Lookup tables and intensity transformations	LUT framework, contrast stretching, image negation, piecewise linear transforms.
4	Acquisition and Geometry	Pinhole camera model and projection	Intrinsic matrix, focal length, principal point, homogeneous coordinates, projection equation with numerical example.
5	Acquisition and Geometry	Lens distortion	Radial and tangential distortion, Brown-Conrady model, calibration, and visual effects of uncorrected distortion.
6	Acquisition and Geometry	Epipolar geometry and stereo vision	Epipolar constraint, fundamental matrix (conceptual), rectification, disparity maps, failure modes.
7	Point Processing	Histogram equalisation and CLAHE	CDF derivation, worked example, CLAHE as adaptive improvement, failure cases.

8	Point Processing	Otsu's thresholding	Within-class variance derivation, global vs adaptive, bimodal assumption and when it breaks.
9	Point Processing	Gamma correction and tone mapping	Power law, perceptual motivation, non-linear encoding, HDR tone mapping at a conceptual level.
10	Area Processing	Convolution kernels	Output dimension equation, stride and padding, Gaussian, Laplacian and Sobel families, parameter sensitivity.
11	Area Processing	Edge detection operators	Sobel, Prewitt, Laplacian of Gaussian, Canny pipeline. Comparison of outputs, failure modes, and parameter sensitivity.
12	Area Processing	Image pyramids and downsampling	Gaussian and Laplacian pyramids, information loss at each level, difference of Gaussians, and multi-resolution motivation.
13	Area Processing	Pooling operations	Max, average, and global average pooling. The equation, spatial effect, what is preserved and what is discarded.
14	Morphology and Segmentation	Structuring elements and morphological sequences	SE shape and size effects, erosion and dilation, opening and closing as sequences, skeleton transform, failure cases.
15	Morphology and Segmentation	Connected components labelling	Two-pass algorithm, 4 vs 8 connectivity, failure modes, practical applications in blob detection and counting.
16	Morphology and Segmentation	Watershed segmentation	Topographic metaphor, markers and over-segmentation, gradient-based watershed, failure modes and alternatives.
17	Feature Detection	HOG descriptor	Gradient computation, cell histograms, block normalisation, full pipeline, parameter choices and failure modes.
18	Feature Detection	Template matching	SSD, SAD, and NCC metrics, sliding window approach, failure under scale and rotation, and when to use each metric.



19	Feature Detection	Panorama stitching and homography	Homography matrix, RANSAC (conceptual), warping and blending, failure modes under parallax and rolling shutter.
20	Motion and Video	Background subtraction, optical flow, and video compression	Frame differencing, running average models, Lucas-Kanade optical flow, brightness constancy assumption, and inter-frame prediction as the basis for video compression. Failure modes across all three.

End of Project Specifications